

Phishing Email Analysis Report

Email File: sample-1034.eml converted to email_body_clean.txt

Analysis Date: 22-Oct-2025

CONVERSION OF EML TO TXT

1 Extract HTML content from the email

If your email (sample-1034.eml) contains HTML formatting, first extract that part:

```
grep -a -A 2000 "<html" sample-1034.eml > email_body.html
```

- `grep -a` → reads the file as text (even if binary parts exist)
 - `-A 2000` → captures 2000 lines *after* `<html>`
 - The output (`> email_body.html`) saves the HTML part separately
-

2 Convert HTML to safe text (no links executed)

Use the lynx command-line browser to render it safely into plain text:

```
lynx -dump -nolist email_body.html > email_body.txt
```

- `lynx -dump` → prints the webpage text content
- `-nolist` → removes numbered hyperlinks
- Output: `email_body.txt` (plain text version)

If you didn't have lynx, you installed it using:

```
sudo apt install lynx -y
```

3 Clean and simplify the text

You can remove extra spaces, control characters or junk formatting:

```
cat email_body.txt | tr -d '\r' | sed '/^$/d' > email_body_clean.txt
```

- `tr -d '\r'` → removes carriage returns
 - `sed '/^$/d'` → removes blank lines
 - Output: **email_body_clean.txt** — clean and readable text body of the email
-

Step 1: Obtain a sample phishing email

Email file link: https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1034.eml

Step 2: Inspect Email Headers

```
root@kali:~/Downloads# grep -i "^Received:" -n sample-1034.eml | nl -ba -v0
0  1:Received: from LV2PR19MB6104.namprd19.prod.outlook.com (::1) by
1  4:Received: from DS7PR06CA0010.namprd06.prod.outlook.com (2603:10b6:8:2a::17) by
2  9:Received: from DM6NAM12FT029.eop-nam12.prod.protection.outlook.com
3 19:Received: from nonkfrgr.co.uk (89.144.9.87) by
```

Here the email is transferred from 4 different servers originating from nonkfrgr.co.uk (89.144.9.87). Let's dig deeper into it.

```
root@kali:~/Downloads# whois 89.144.9.87
% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See https://docs.db.ripe.net/terms-conditions.html
%
% Note: this output has been filtered.
%       To receive output for a database update, use the "-B" flag.
%
% Information related to '89.144.9.0 - 89.144.9.255'
different servers originating from
% Abuse contact for '89.144.9.0 - 89.144.9.255' is 'abuse@ghostnet.de'
%
inetnum:        89.144.9.0 - 89.144.9.255
netname:        DE-GHOSTNET-20250221
country:        DE
org:            ORG-GG3-RIPE
admin-c:        GN-RIPE
tech-c:         GN-RIPE
status:         ASSIGNED PA
mnt-by:         GHOSTNET-MNT
mnt-lower:      GHOSTNET-MNT
mnt-routes:     GHOSTNET-MNT
created:        2025-03-10T13:46:10Z
last-modified:  2025-03-10T13:46:10Z
source:         RIPE # Filtered

organisation:   ORG-GG3-RIPE
org-name:       GHOSTnet GmbH
country:        DE
org-type:       LIR
address:        Am Dachsbau 17
address:        65812
```

```
address:      Bad Soden a.Ts.
address:      GERMANY
phone:        +49 6172 185025
fax-no:       +49 6172 185029
admin-c:      GNSB-RIPE
admin-c:      GN-RIPE
admin-c:      GNSG-RIPE
tech-c:       GN-RIPE
abuse-c:      GN-RIPE
mnt-ref:      RIPE-NCC-HM-MNT
mnt-ref:      GHOSTNET-MNT
mnt-by:       RIPE-NCC-HM-MNT
mnt-by:       GHOSTNET-MNT
created:      2004-04-17T11:07:35Z
last-modified: 2020-12-16T13:16:24Z
source:       RIPE # Filtered

role:         GHOSTnet GmbH
admin-c:      GNSG-RIPE
tech-c:       GNSG-RIPE
address:      Am Dachsbau 17
address:      65812 Bad Soden a. Ts.
address:      Deutschland
phone:        +49 6172 185025
fax-no:       +49 6172 185029
nic-hdl:      GN-RIPE
abuse-mailbox: abuse@ghostnet.de
mnt-by:       GHOSTNET-MNT
created:      2003-04-17T02:22:16Z
last-modified: 2017-11-10T09:36:32Z
source:       RIPE # Filtered
```

% Information related to '89.144.9.0/24AS12586'

```
route:        89.144.9.0/24
origin:       AS12586
mnt-by:       GHOSTNET-MNT
created:      2025-03-10T13:46:10Z
last-modified: 2025-03-10T13:46:10Z
source:       RIPE
# Objects that do not appear in a standard
```

% This query was served by the RIPE Database Query Service version 1.119 (ABERDEEN)

Step 3: Checking email headers for discrepancies using online header analyser

Analysed using Message Header analyser

Message Header Analyzer

Insert the message header you would like to analyze

Received: from LV2PR19MB6104.namprd19.prod.outlook.com (::1) by MN0PR19MB6312.namprd19.prod.outlook.com with HTTPS; Thu, 3 Aug 2023 03:57:53 +0000

Received: from DS7PR06CA0010.namprd06.prod.outlook.com (2603:10b6:8:2a::17) by LV2PR19MB6104.namprd19.prod.outlook.com (2603:10b6:408:171::16) with Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6631.47; Thu, 3 Aug 2023 03:57:52 +0000

Received: from DM6NAM12FT029.eop-nam12.prod.protection.outlook.com (2603:10b6:8:2a:cafe::5f) by DS7PR06CA0010.outlook.office365.com (2603:10b6:8:2a::17) with Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6652.20 via Frontend Transport; Thu, 3 Aug 2023 03:57:52 +0000

Authentication-Results: spf=none (sender IP is 89.144.9.87) smtp.mailfrom=nonkfrgr.co.uk; dkim=none (message not signed)

Analyze headers

Clear

Copy

[Submit feedback on github](#)

Summary

Subject

Microsoft account unusual signin activity

Message Id

<996dcdb8-3d55-4865-93d2-d8471d7db546@DM6NAM12FT029.eop-nam12.prod.protection.outlook.com>

Creation time

Thu, 3 Aug 2023 03:57:52 +0000 (Delivered after 1 second)

From

Microsoft account team _<no-reply@access-accsecurity.com>

Reply to

solutionteamrecognizd03@gmail.com

To

phishing@pot

Received headers

Hop:	Submitting host	Receiving host	Time	Delay	Type
1	nonkfrgr.co.uk (89.144.9.87)	DM6NAM12FT029.mail.protection.outlook.com (10.13.178.133)	8/3/2023 9:27:52 AM		Microsoft SMTP Server
2	DM6NAM12FT029.eop-nam12.prod.protection.outlook.com (2603:10b6:8:2a:cafe::5f)	DS7PR06CA0010.outlook.office365.com (2603:10b6:8:2a::17)	8/3/2023 9:27:52 AM	0 seconds	Microsoft SMTP Server (version=TLS cipher=TLS_ECDHE_RSA_WITH_A
3	DS7PR06CA0010.namprd06.prod.outlook.com (2603:10b6:8:2a::17)	LV2PR19MB6104.namprd19.prod.outlook.com (2603:10b6:408:171::16)	8/3/2023 9:27:52 AM	0 seconds	Microsoft SMTP Server (version=TLS cipher=TLS_ECDHE_RSA_WITH_A
4	LV2PR19MB6104.namprd19.prod.outlook.com (::1)	MN0PR19MB6312.namprd19.prod.outlook.com	8/3/2023 9:27:53 AM	1 second	HTTPS

Microsoft Antispam Header

Bulk Complaint Level [6](#)
Source header [BCL:6](#);

Other headers

#	Header	Value
1	Authentication-Results	spf=none (sender IP is 89.144.9.87) smtp.mailfrom=nonkfrgr.co.uk; dkim=none (message not signed) header.d=none;dmarc=per merror action=none header.from=access-accsecurity.com;
2	Received-SPF	None (protection.outlook.com: nonkfrgr.co.uk does not designate permitted sender hosts)
3	X-IncomingTopHeaderMarker	OriginalChecksum:2AD9067109479DC9991A43AD0BAFA157A1EB87646DD379C1957A6762D9DD58F;UpperCasedChecksum: C5853EF871EC39165D7DB0546167D17A3FFFFF3C5DF2B707D539A4256DC12907;SizeAsReceived:396;Count:12
4	Content-Length	18708448
5	Content-Length	1821750
6	Content-Type	text/html; charset="UTF-8"
7	Content-Transfer-Encoding	8bit
8	X-IncomingHeaderCount	12
9	Return-Path	bounce@nonkfrgr.co.uk
10	X-MS-Exchange-Organization-ExpirationStartTime	03 Aug 2023 03:57:52.8482 (UTC)
11	X-MS-Exchange-Organization-ExpirationStartTimeReason	OriginalSubmit
12	X-MS-Exchange-Organization-ExpirationInterval	1:00:00:00.0000000
13	X-MS-Exchange-Organization-ExpirationIntervalReason	OriginalSubmit
14	X-MS-Exchange-Organization-Network-Message-Id	fa529702-084b-46b8-503a-08db93d5cfe8
15	X-EOPAttributedMessage	0
16	X-EOPTenantAttributedMessage	84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa:0
17	X-MS-Exchange-Organization-MessageDirectionality	Incoming
18	X-MS-PublicTrafficType	Email
19	X-MS-TrafficTypeDiagnostic	DM6NAM12FT029:EE_JLV2PR19MB6104:EE_JMN0PR19MB6312:EE_
20	X-MS-Exchange-Organization-AuthSource	DM6NAM12FT029.eop-nam12.prod.protection.outlook.com
21	X-MS-Exchange-Organization-AuthAs	Anonymous
22	X-MS-UserLastLogonTime	8/3/2023 3:19:25 AM
23	X-MS-Office365-Filtering-Correlation-Id	fa529702-084b-46b8-503a-08db93d5cfe8

24	X-MS-Exchange-EOPDirect	true
25	X-Sender-IP	89.144.9.87
26	X-SID-PRA	NO-REPLY@ACCESS-ACCSECURITY.COM
27	X-SID-Result	NONE
28	X-MS-Exchange-Organization-PCL	2
29	X-MS-Exchange-Organization-SCL	5
30	X-MS-Exchange-CrossTenant-OriginalArrivalTime	03 Aug 2023 03:57:52.6764 (UTC)
31	X-MS-Exchange-CrossTenant-Network-Message-Id	fa529702-084b-46b8-503a-08db93d5cfe8
32	X-MS-Exchange-CrossTenant-Id	84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa
33	X-MS-Exchange-CrossTenant-AuthSource	DM6NAM12FT029.eop-nam12.prod.protection.outlook.com
34	X-MS-Exchange-CrossTenant-AuthAs	Anonymous
35	X-MS-Exchange-CrossTenant-FromEntityHeader	Internet
36	X-MS-Exchange-CrossTenant-RMS-PersistedConsumerOrg	00000000-0000-0000-0000-000000000000
37	X-MS-Exchange-Transport-CrossTenantHeadersStamped	LV2PR19MB6104
38	X-MS-Exchange-Transport-EndToEndLatency	00:00:00.9978232
39	X-MS-Exchange-Processed-By-BccFoldering	15.20.6631.046
40	X-Message-Flag	Flag
41	Importance	high
42	X-Priority	1
43	X-Microsoft-Antispam-Mailbox-Delivery	abwl:0;wl:1;pcwl:1;kl:0;dwl:0;dkl:0;rwf:0;ucf:0;jmr:0;ex:0;auth:0;dest:1;OFR:TrustedSenderList;ENG;(5062000305)(920221119095)(900001117)(920221120095)(90002001)(91000020)(91036095)(91040095)(9050020)(9055020)(9100338)(944500132)(2008001134)(2008121020)(4810010)(4910033)(8820095)(9710001)(9610025)(9525003)(10120022)(9439006)(9310011)(9220031)(120001);
44	X-Message-Delivery	Vj0xLJE7dXM9MDtsPTA7YT0wO0Q9MTiHRD0yO1NDTD0tMQ==
45	X-Microsoft-Antispam-Message-Info	6ZfWXPev6/ffZl/kzu8p61jja4rzlb+haQsx7U0KfO2Jp8CKxMEIstffv+NDufEI31xnvGhAKMFPCNbqR2P2kdqw20QxFD6X4pudWvPX T1mMOH+XaQcJS6k7Y0kktCsVFRgJ+ylEBVYeX7ZKtfDe3a1567GibnYe5zhvrxFlplUbJIUaNFdURfsHXLJ5Gnq05al7M8YC9GSB mf3U6e5/dW02sWL7cka22BWKInMeBbGvTCQ2yrkkeVvECroJSJdlMTiy4XdMVkPE4/8COROp0uKVvIYof1cqK/1ggd85d+R2Y 2dix5UaruPTJYz8SlnUG6eMFadl+CCH3KcGhxxG8MIRy3QeMZO3ZGz/D1FnFyMU+OPHTZgY096SFkznRXCYL7j22j6Wkk7AF pJpBCeZP5HmV8aAb8xuNceNhgVnLH2zmkgWu0pLv8xtPaSSDQCDZAcSIBYdJk3F+VKG5aSmI6zK0+7gz1P9dobRSPHn9xZa vMMU9oyQud89CRp9vvOXWldQPJ+owDkFXkRiHwsAongXbnzA23gl5MpR4nTEPUiXNzUJR2gaHpJAsGkmmtd8FKC9XV3CKY byV3RZu8GH4MMCMfKww4i53xrwkATZfdWcRODWFN/S/h51fuArz92wacoC/XDU7+cEGFwhls+ixNYA8zkwEDxFMKPevrZYMcd wdoFBNijVm6DGC+ZWf5s/x5TXfGIApRFFCxxnzAGWb7qc9rp/nEiW3ae1a/4ZpylqwkcdTZg92ldzUHqFPkHTUzFSVW/GzNIGX WC0Lx8RzScjmd5eOHZRPrAhb1GzADDDGOFDsemLrpH3G5T3Uygx4pJtjZGZhySe33uuuBEFeyAiQXwJGnRoYzdSydETVUUU mjN8FoZn0wN48ttXMedeCKWeiSdR3+tsVOC.JjqNoR8PwiKPrUfjcqyIKQGeTvRKJD4HJzz50VUMa6vftbptLDLD1xWE+5nAzP5gl b9+fgMCp5Vh7GsFNeVfzoLkug21FdFa038mN2cSBtNRI5DpjinTomEUcAvuF0D3ZyjpgSSGqwx5sv3ePQvfln5r8702dArGLTZ1 Pj2ckk0H5PQ2blOd03NctP8HxUNAexxm7LoyJNjErkGJZcAmuTJfnz90wfpPfmbQvRNhRHF8uVcvT/7BBNhlz7SsAb85UjppOD02a cLOml+gp1EZQTDBeYZCcaYjh+EBDZ4phGwuAvGwkpkrswm2+TIVHQIHQJ2DX6bhgyoWNIOojWTzpYEXRPYwAYAwjWu3Bw

Analysed using **whatismyip.com**

Tool URL: <https://www.whatismyip.com/email-header-analyzer/>

Email Source IP Info

The Email Source IP Address is 103.225.77.255

The Email Source Hostname is 103.225.77.255

ASN: -

City: New Delhi

State/Region: Delhi

Country: India

Postal Code: 110001

ISP: Adsizzler Media Pvt Ltd

Step 4: Identify Suspicious Links or Attachments

Command:

```
grep -iE 'sign[- ]?in|account|report|opt out|click here|unusual|Russia|IP|device' human_text.txt
```

Output:

```
root@kali:~/Downloads# grep -iE 'sign[- ]?in|account|report|opt out|click here|unusual|Russia|IP|device' human_text.txt
Microsoft account
Unusual sign.in activity
We detected something unusual about a recent sign-in to the Microsoft
account phishing@pot.
Sign-in details
Country/region: Russia/Moscow
IP address: 103.225.77.255
A user from Russia/Moscow just logged into your account from a new
device, If this wasn't you, please report the user. If this was you,
Report The User
To opt out or change where you receive security notifications, click
The Microsoft account team
```

Analysis: Headers show multiple unusual indicators — Return-Path domain mismatch, anonymous authentication and message routing from non-official sources.

Step 5: Check for Urgent or Threatening Word

Command:

```
grep -iE 'click|report|opt out|verify' human_text.txt
```

Output:

```
root@kali:~/Downloads# grep -iE 'click|report|opt out|verify' human_text.txt
device, If this wasn't you, please report the user. If this was you,
Report The User
To opt out or change where you receive security notifications, click
```

Analysis: Urgent calls to action and suspicious links are indicators of phishing.

Step 6: Finding any mismatching URLs

Command:

```
grep -iE "<a|<img|<form" sample-1034.eml
```

Output:

```
root@kali:~/Downloads# grep -iE "<a|<img|<form" sample-1034.eml
<td id="i3" style="padding: 25px 0px 0px; font-size: 14px; font-family: 'Segoe UI', Tahoma, Verdana, Arial, sans-serif; color: #2a2a2a; height: 18px; width: 696px;">We detected something unusual about a recent sign-in to the Microsoft account <a id="iAccount" class="link" dir="ltr" style="color: #2672ec; text-decoration: none;" href="mailto:solutionteamrecognizd03@gmail.com?&cc=solutionteamrecognizd03@gmail.com&Subject=Report+The+User">phishing@pot</a></td>
<td style="background-color: #2672ec; min-width: 50px; padding: 5px 20px 5px 20px;" bgcolor="#2672ec"><a id="i11" style="font-family: 'Segoe UI Semibold', 'Segoe UI Bold', 'Segoe UI', 'Helvetica Neue Medium', Arial, sans-serif; font-size: 14px; text-align: center; text-decoration: none; font-weight: 600; letter-spacing: 0.02em; color: #fff;" href="mailto:solutionteamrecognizd03@gmail.com?&cc=solutionteamrecognizd03@gmail.com&subject=unusual signin activity&body=Report The User">Report The User</a></td>
<td id="i12" style="padding: 25px 0px 0px; font-family: 'Segoe UI', Tahoma, Verdana, Arial, sans-serif; font-size: 14px; color: #2a2a2a; height: 18px; width: 696px;">To opt out or change where you receive security notifications, <a id="iLink5" class="link" style="color: #2672ec; text-decoration: none;" href="mailto:solutionteamrecognizd03@gmail.com?&cc=solutionteamrecognizd03@gmail.com&Subject=Unsubscribe+me">click here</a></td>

```

Analysis: Here the line

```
<imgsrc="http://thebandalisty.com/track/o40209FNaIZ18708448XKuS1821750Wgf33736QaXI176">
```

Even if it's an image, that's still a tracking or malicious link 🚩

Step 7: Verify Spelling or Grammar Errors

Command:

```
aspell list < email_body_clean.txt | sort -u
```

Purpose: Identify unusual or suspicious words in the email that do not appear in a standard dictionary.

Output (excerpt):

```
Vmo
VnMjFGZEZhMDM
Vp
VtTHJwSDNHNvQzVXlneDRwSnRqWkdaaHlTZTMzdXVpQkVG
WEVqOEFpMTh
WFBBAk
WGLMRG
WHlpQXRITzlBZ
WOGFBYjh
WTBra
WUM
WaEFxZnhDT
Wi
WkZucUhhbHY
WlA
WlNjam
WmhZULJhdVJYtlg
WnM
YNVdrOVM
YTRSa
YTlPaU
YVntSTZ
YkRzdEt
YlVvWUQ
YnZnTjRDTkJX
YxSTVPemRGRTV
ZCTmpqVm
ZCtSMlkyZGl
ZDg
ZEc
ZEgvMDJwSDNVmml
ZFd
ZG
ZGUmdKK
ZGs
```

Analysis: Several words are unusual and seem to have errors.

Step 8: Summarize Phishing Indicators

Checkpoint	Finding	Status	Severity
1. Sample email obtained	File sample-1034.eml collected and analyzed safely in a sandbox	✅ Legitimate procedure	● Low
2. Examine sender address	From: Microsoft account team <no-reply@access-accsecurity.com> — fake Microsoft lookalike domain	🚩 Spoofed / Fake sender	● High
2a. Reply-To mismatch	Reply-To: solutionteamrecognizd03@gmail.com — attacker-controlled Gmail address	🚩 Redirects responses to attacker	● High
3. Header analysis	Origin IP 89.144.9.87 (non-Microsoft server, Ghostnet GmbH Germany), Return-Path bounce@nonkfrgr.co.uk	🚩 Spoofed mail routing	● High
3a. SPF/DKIM/DMARC	No DKIM/DMARC alignment; SPF validation failed; X-SID-Result: NONE	🚩 Authentication failed	● High
4. Suspicious links / attachments	Hidden tracking pixel — http://thebandalisty.com/track/... → NXDOMAIN (domain does not exist)	🚩 Malicious tracking / phishing indicator	● Medium
5. Urgent or threatening language	“Unusual sign.in activity”, “Report The User”, “click here” — urgency tactic	⚠️ Social engineering	● Medium
6. Mismatched URLs	Displayed brand: Microsoft, but real domain: thebandalisty.com	🚩 Misleading / phishing redirect	● High
7. Grammar & spelling	Errors like “sign.in”, inconsistent capitalization, awkward phrasing	⚠️ Common phishing trait	● Low
8. Final assessment	Multiple red flags: spoofed sender, fake headers, phishing links, urgency, grammar errors	❌ Phishing confirmed	● Critical

Overall Severity: Critical

Recommendation: Do **not click any links**, block the sender/domain and report phishing to Microsoft.



MITIGATION

1. Block the sender in your email client.
2. Report phishing to Microsoft via <https://reportphishing.microsoft.com>
3. Verify account activity directly on Microsoft's official site.
4. Avoid downloading attachments or clicking links from suspicious emails.

Dheepan Ganesan